assuming that such a number exists. If no such number exists, then $L(\alpha) = \infty$.

Thus we have $L(\frac{1+\sqrt{5}}{2}) = \sqrt{5}$. Furthermore, if we let $\phi = \frac{1+\sqrt{5}}{2}$ and choose integers a, b, c, d such that $ad - bc = \pm 1$, then $L\left(\frac{a\phi+b}{c\phi+d}\right) = \sqrt{5}$ as well. However, for any irrational number α not of this form, $L(\alpha) \geq \sqrt{8}$. The set of numbers of the form $L(\alpha)$ for some irrational number α is called the *Lagrange spectrum.*

Good places to read more about this and related mathematics are [SS07, Chapter 4] and [Bak19].

After that, we can wonder about increasing the exponent in the denominator. What does it say about a number if it can be approximated to within the cube of the denominator, say, or a higher power? There's a beautiful theorem due to Liouville on that:

Theorem 4.14 (Liouville). *Let α be an irrational number that is a root of a degree-n polynomial with integer coefficients. Then there exists a constant $C > 0$ (depending on α) such that if $\frac{p}{q}$ is any rational number, then*

$$\left|\alpha - \frac{p}{q}\right| \geq \frac{C}{q^n}.$$

But there are some numbers that can be approximated to within $\frac{C}{q^n}$ for *any* positive integer n. It follows, then, that such numbers cannot be roots of any nonzero polynomial with integer coefficients, i.e. they are *transcendental.* A popular example is

$$\alpha = \sum_{n=1}^{\infty} 10^{-n!} = .11000100000000000000000001000\ldots,$$

where the 1's are in the factorial positions. Liouville's Theorem is the starting point of the fascinating subject of transcendental number theory. A good place to start learning about it is [BT04].

4.4 Problems

(1) Suppose there are nine points (a_i, b_i, c_i) with $1 \leq i \leq 9$ in three-dimensional Euclidean space, also known as $\mathbb{R}^3$, all of whose coordinates are integers (i.e. a_i, b_i, c_i are integers). Show that there are two points among these nine such that the midpoint of the segment connecting them has integer coordinates. For instance, if two of the points are $(1, 2, 3)$ and $(4, 6, 8)$, then their midpoint is $(\frac{5}{2}, 4, \frac{11}{2})$. So

this midpoint does not have integer coordinates, but some pair must if there are 9 points.

(2) Let S be a set consisting of 51 positive integers from 1 to 100. Prove that S must contain two consecutive integers.

(3) Find, with proof, the maximum number of knights one can place on a chessboard such that no two of them attack each other. Do the same for bishops.

(4) Prove that if S is any subset consisting of 10 distinct integers from 1 to 100, there are two nonempty disjoint subsets $T, U \subseteq S$, such that the sum of the integers in T is equal to the sum of the integers in U.

(5) Prove that if n is any positive integer, there is a positive multiple of n whose decimal expansion contains only 0's and 1's. For instance, 1001 is a multiple of 7 consisting of only 0's and 1's.

(6) Prove that among any set S of $n \geq 2$ people, there are two of them who have the same number of friends in S. We assume the friendship is mutual: if person A is friends with person B, then person B is also friends with person A.

(7) Prove that, given any five points on the surface of a sphere, there is some closed hemisphere (i.e. including the boundary) that contains at least four of them.

(8) Let $a_0, a_1, \ldots, a_n$ be any integers. Prove that

$$\prod_{0 \leq i < j \leq n} (a_i - a_j)$$

is a multiple of $n!$.

(9) (a) Prove that for any positive integer k, we can find k consecutive positive integers that are all composite.

(b) Prove that the infinite sum

$$\sum_{p \text{ prime}} \frac{1}{2^p}$$

is an irrational number.

Alexander

T+P Week 4: Pigeonhole Principle 10/13-18/26

4.1: Consider the point (a_1, b_1, c_1). If each value were "even" or "odd", there are $2 \cdot 2 \cdot 2 = 2^3 = 8$ possible points in this form. Each midpoint $\left(\frac{a_1+a_2}{2}, \frac{b_1+b_2}{2}, \frac{c_1+c_2}{2}\right)$ can only have integer coordinates if a_1+a_2, b_1+b_2, and c_1+c_2 are even. That happens when both a's, b's, and c's are even or odd. We can make 8 points (a_i, b_i, c_i) such that their parity never completely lines up, but by Pigeonhole, a ninth point must match with at least 1 other, having a midpoint with int. coords.

4.2: Pigeons: numbers in S

Holes: $(1,2), (3,4), \ldots, (99, 100)$.

There are 50 holes and 51 pigeons, so there must be 2 pigeons in a hole. Each hole has 2 consecutive numbers, so there exists a pair between 1 and 100. Q.E.D.

4.3: Notice that a knight can only move from white -> black or vice versa, not white -> white or black -> black. Thus, if we place a knight on <u>every white (or black) square</u>, none will attack each other. That adds up to 32 knights.

Proof. Pigeons: 33 knights

Holes: 32 pairs of attacking squares (1 black 1 white)

Each knight can be placed in 32 holes, and 1 knight will be stuck attacking another.

For bishops, place 8 in one row/column and 6 in the other (avoiding corners).

Proof. Pigeons: 15 Bishops

Holes: 13 rows from ↙, plus 2 corners as one "group" that attack eachother, so 14.

By Pigeonhole, 2 bishops must be attacking, so 14 max. ■

4.4: Note the highest possible sum is 955 (for a subset) and the lowest is 1. (cont'd.)

*-1 is to remove the empty set, or $\{\emptyset\}$.

4.4 (cont'd)... Also note that the number of subsets possible is $2^{10}-1^{*}=1023$, or 10 choices of "yes" or "No" in S. For distinct (not disjoint) sets, at least. In this case, 1023 pigeons don't fit in 955 holes, so at least two share a sum. But what if T and U share a value in S? It turns out that we can "delete" the terms T and U share while keeping their sums the same, boiling all possible subsets down to the 1023 (in the case of equal sums). This deletion process can never produce the empty set ($\emptyset$) because $T \neq U$ according to the theorem. Thus, the Pigeonhole Principle still holds, and $\sum_{x \in T} x = \sum_{x \in U} x$ is true for at least one pair of T and U.

4.5: Say that S(m) = a number with m digits, all 1s, so S(0) = 0, S(1) = 1, S(2) = 11, S(3) = 111...1111, 11111 (cont.)...

4.5: (cont.) In the list $(S(1), S(2), \ldots, S(n), S(n+1))$, each $S(m)$ has a remainder when divided by n that is $\leq n-1$ (and ≥ 1). This means n remainders in all, but with $n+1$ terms at least 2 must have the same remainder. (Pigeonhole!) Subtracting these two will yield an exact multiple of n, which is a number made of just 0's and 1's.

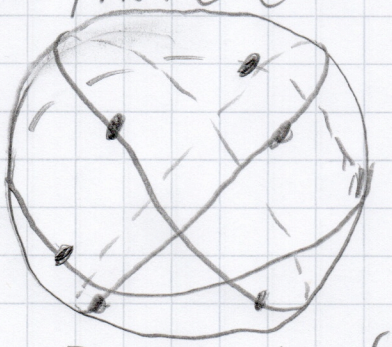
4.6: Two Cases:

Nobody has 0 friends. Everyone can have $1 \leftrightarrow (n-1)$ total friends, or $n-1$ options.

At least 1 person is alone. Everyone else can have $0 \leftrightarrow (n-2)$ total friends, or $n-1$ options, because nobody can befriend the loner.

So everyone has $n-1$ options for friend count, and there are n people, so at least 2 must be friends by Pigeonhole. ∎

4.7: Any 2 points define a great (hemisphere) circle on a sphere (as they do a line in the plane).



Call these 2 points (P_1, P_2). The other points (P_3, P_4, P_5) can go in either hemisphere denoted by P_1 and P_2. That's three pigeons (points), and two holes (hemispheres). So at least 2 must be in one hemisphere. Because the boundary is included, 4 points (at least) must be in one hemisphere. Kudos to the P_5 point if it joins, but the problem only asked for 4 points.

4.8: Pigeons: $a_0, a_1, \ldots, a_n$.

Holes: Remainders $0, 1, \ldots, K-1$ for an integer K selected from 2 to n (1 is trivial). Here, at least two of $a_0 \ldots a_n$ must share the same remainder when $\div k$. (cont'd. on next page)

*The ones with the same remainder ÷ k.

4.8 (cont'd.)... In terms of $\prod_{0\le i<j\le n}(a_i - a_j)$, at least one $(a_i - a_j)$ must have both a_i's selected from the list,* and their difference is a multiple of k. Because k can be from 2 to n, a multiple of 2, 3, ..., n exists at least once in $\prod_{0\le i<j\le n}(a_i - a_j)$. The product $2 \cdot 3 \cdot \ldots \cdot n$ is just equal to n!, so n! exists as a factor in the big product. ∎

4.9: a) These k numbers can be expressed as: $k!+2, k!+3, k!+4, \ldots, k!+k$. Each are divisible by 2, 3, ..., k, so all are composite. ∎

b) This sum can be expressed in binary as:

$0.01101010001010001010\ldots_2$

or the sum $\frac{1}{2^2} = 0.01_2 + \frac{1}{2^3} = 0.001_2 + \frac{1}{2^5} = 0.00001\ldots$ such that each 1 is in the position of a prime number (but not base 10) when counting from 1 after the decimal point. This must be irrational because primes are infinite and the composites between them are too (qa), so no repetition exists. ∎